

particular word following another will lead the computer towards finding a match. Finally, the need is that you should be able to interact with your personal networks simply by talking and listening to these.

Apart from human-to-machine communication, reliable machine-to-machine communication is also important, especially if the personal network is to be able to interact effortlessly with other people's personal networks. Channels of communication comprise chiefly the physical links between machines like home LANs, infrared links among devices located in very close proximity, short-range radio links used in cordless phones, long-range radio links as in mobile phones and cable or high-speed connections to the rest of the world.

Wireless communication techniques form the hub to the personal network mobile computing. While for in-room communication infrared systems offer links of up to 4Mbps across a distance of about one metre, wide-area usage demands radio links.

Complexities of wireless communications are far greater than those posed by wired links. Key communication parameters such as transmission speed, error rate, reliability and delay can change considerably and rapidly during the course of a single wireless communication. This instability is driven by the variability of radio frequency noise and the signal attenuation that is caused by natural phenomenon such as storms or due to interference from various electronic devices.

Mobility further brings in more uncertainty. For example, radio-wave propagation is abruptly inter-



Fig. 4: A prototype of a new speech-controlled nutrition-logging system allows users to verbally describe the contents of a meal. The system then parses the description and automatically retrieves pertinent nutritional data (Image courtesy: <http://news.mit.edu>)



Fig. 5: Access to personal network devices will be increasingly protected by biometrics (Image courtesy: www.scoop.it)

rupted when a skyscraper, vehicle or some other large object comes between the transmitting and receiving devices. Therefore transmission speeds available in wireless communications tend to be much lower vis-a-vis wired links.

Security issues

Access to personal network devices will be increasingly protected by biometrics—use of physiological features such as pattern of a voice, fingerprint or iris to permit activation. Biometrics are unique to an individual. For example, no other human being on Earth has the same fingerprint as yours and, hence, your thumb or finger swipe allows you and only you to access your laptop.

Traditional protection such as passwords will exist but only to supplement biometrics. However, public-key cryptography—a mathematical system that will be used to secure owner control over the personal network and allow for intra- as well as inter-network privacy—will form the hub of the security system. Encryption systems known as RSA and PGP are examples of public-key cryptography.

Public-key cryptography works by assigning each person two different keys, which really are two numbers linked with each other. The two keys are several digits in length and next to impossible for most humans to memorize. These are generally saved on the computer's hard drive or on a disk. The public key is accessible to anyone who wants to send you a secure message. The private key, however, is only available with you. Public-key cryptography guards with encryption and ensures privacy. A sender can en-

crypt a message to you that could be an account number or a debit card number using your public key. However, this message can be decoded only by using both the public and the private key, and only you have the private key.

Other issues

Apart from security, another critical aspect is software. Programs need to be such that these cater to multiple computers and operating systems. Software will need to be written in a language like Java that can translate commands into readable instructions and is device agnostic.

Maybe one of the biggest challenges to the development of the personal network (as it is also for